

Vector Database Architecture and Latency Benchmarks

System Architecture & Performance Overview

This report analyzes the retrieval latency of various vector indexing backends for high-throughput RAG workloads. Our enterprise architecture integrates Postgres with pgvector using HNSW (Hierarchical Navigable Small World) indexes to balance search latency and strict multi-tenant data isolation.

Key benchmark findings:

- pgvector HNSW achieves 12ms average latency for 1024-dimensional embeddings
- FAISS in-memory index reaches 8ms latency but requires manual index sharding
- Managed cloud vector databases show 28ms to 45ms latency under similar load

Vector Database Query Latency Comparison (1024-dim, Top-K=10)

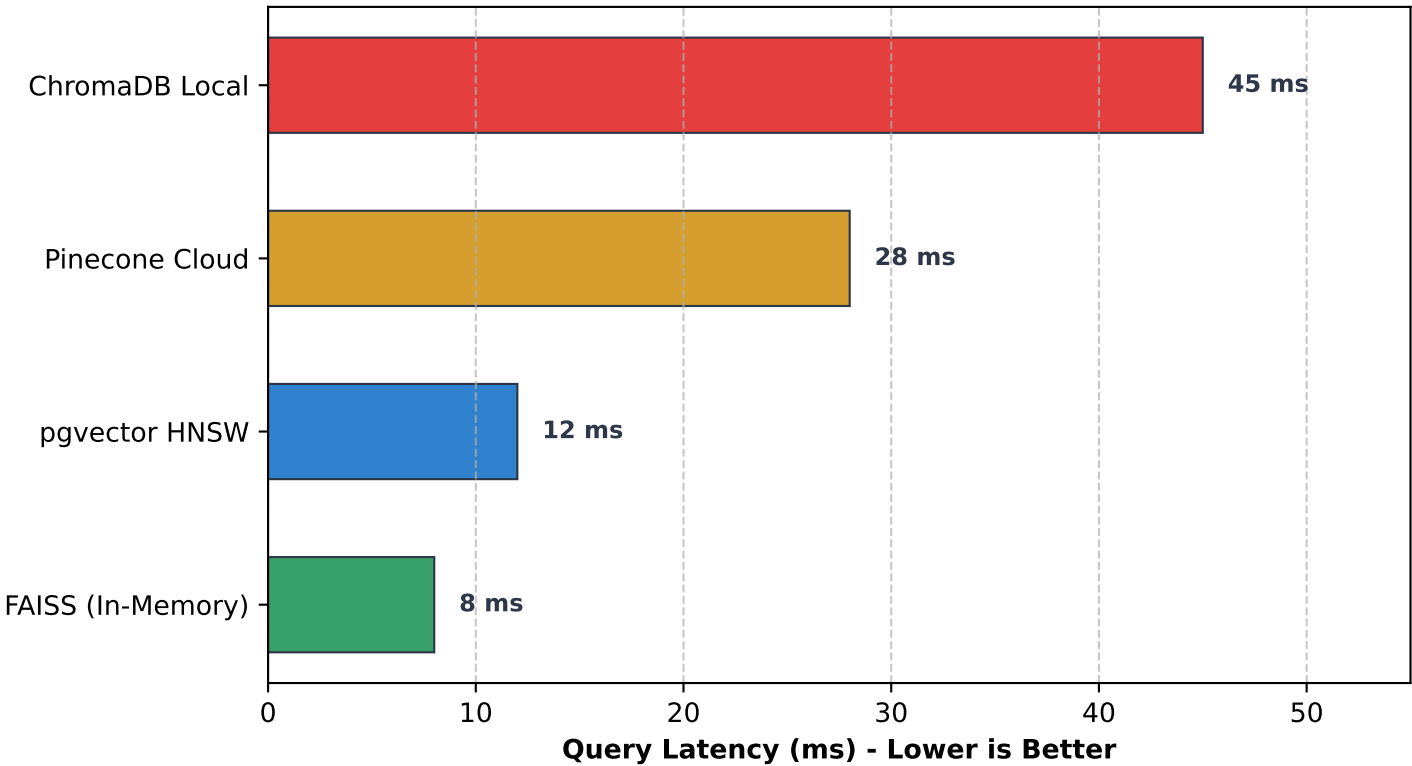


Chart Analysis & Visual Summary:

As depicted in the latency comparison chart above, pgvector HNSW provides the optimal balance of speed (12ms) and transactional ACID guarantees while keeping all client data and embedding vectors securely hosted on-premise without external API egress.